gps-gis-devops-project

DevOps portfolio: GPS/GIS microservices with full CI/CD on AWS EKS

GPS/GIS DevOps Portfolio Project

A full-stack GPS/GIS application tracking capital cities and geographical points of interest around the world, built as a DevOps portfolio project demonstrating the complete modern DevOps toolchain.

🌍 Live Features

- Interactive world map with clickable markers for capital cities and special interest locations
- Expandable location cards showing coordinates and plain-language remarks
- Active/Passive toggle — show or hide any location on the map instantly
- Alphabetical sort and real-time search across all tracked points
- Any geographical point on Earth can be marked, tracked, and given plain-language remarks — demonstrated by a dam in Iran included as a real-world example

🛠 Tech Stack

Layer	Technology
Backend	Python FastAPI + PostgreSQL
Frontend	React + Vite + Leaflet Maps
Containerization	Docker + Docker Compose
CI/CD	Jenkins Pipeline
Orchestration	Kubernetes (minikube + AWS EKS)
Cloud	AWS EKS + ECR
Infrastructure as Code	Terraform

📊 Monitoring

Prometheus and Grafana deployed via Helm into a dedicated `monitoring` namespace.
Access Grafana dashboard:

```
kubectl port-forward -n monitoring svc/monitoring-grafana 3000:80
# Open http://localhost:3000
```

Commit with:

🏗 Architecture

```
Developer → GitHub → Jenkins CI/CD → Docker Images → Kubernetes → AWS EKS
                                                ↓
                                React Frontend + FastAPI Backend + PostgreSQL
```

🚀 Quick Start

Run with Docker Compose

```
cd docker
docker-compose up
# Open http://localhost:5173
```

Run with Kubernetes (minikube)

```
minikube start --driver=docker
eval $(minikube docker-env)
docker build -t gps-backend:latest ./backend
docker build -t gps-frontend:latest ./frontend
kubectl apply -f kubernetes/
minikube service frontend-service --url
```

📊 API Endpoints

Method	Endpoint	Description
GET	/health	Health check
GET	/api/capitals	Get all cities
GET	/api/capitals/{id}	Get one city
POST	/api/capitals	Create city
PUT	/api/capitals/{id}	Update city
DELETE	/api/capitals/{id}	Delete city

🧪 Tests

```
cd backend
pytest tests/ -v
```

7/7 tests passing ☑

🗺 Future Roadmap

Coming Next

Enhancement	Description
Argo CD	Full GitOps CD pipeline with auto-sync from Git
Helm Charts	Package Kubernetes manifests for reusable deployments
ConfigMaps & Secrets	Separate configuration from code, no hardcoded values
Grafana + Prometheus	Real-time monitoring dashboards and alerting
Image Scanning	Trivy security scanning integrated into Jenkins pipeline
RBAC	Role-based access control with least-privilege principles
Network Policies	Pod-level network security and traffic control

> *"A good DevOps engineer never stops improving the pipeline."*

👤 Author

DevOps-ily — Built from scratch in under 2 weeks, under missile attacks, because nothing stops a DevOps engineer. 加油! 💚